

UICT — Uganda
Department of Computer Science

CSC 121

Computer Hardware & Maintenance

Hardware, Components & Troubleshooting

Ground to Mastery Study Notes

Bravehart
DCS Year 1, Semester 2
Academic Year 2025/2026

Assessment: Coursework 40% | Final Exam 60%

Assessment Breakdown

Component	Weight	Details
Continuous Assessment (Coursework)	40%	Tests, assignments, practicals
Final Examination	60%	Written final exam at semester end

Minimum pass mark: 50% overall

You must pass BOTH components to pass the course unit.

Unit 1: Overview of Computer Systems

A computer system is an integrated set of hardware and software components that work together to process data and produce useful information. Understanding the interaction between these components is fundamental to hardware maintenance.

Term	Definition
Hardware	Physical, tangible components of a computer system
Software	Programs and operating systems that instruct hardware what to do
Firmware	Permanent software stored in ROM that controls hardware at the lowest level
Input device	Hardware used to enter data into the computer (keyboard, mouse, scanner)
Output device	Hardware used to present processed data (monitor, printer, speakers)
Processing unit	The CPU — performs all calculations and data processing
Storage device	Hardware that stores data temporarily (RAM) or permanently (HDD, SSD)

Classification of Computers

- Supercomputer — extremely powerful; used for scientific research and weather forecasting
- Mainframe — large, high-capacity computers used by banks and governments for bulk data processing
- Minicomputer — mid-range computers for small businesses
- Microcomputer (PC) — personal computers for individual use; includes desktops, laptops, tablets

EXAM TIP: Know the Von Neumann architecture: Input → Processing (CPU + Memory) → Output → Storage. The CPU fetches, decodes, and executes instructions in a cycle called the Fetch-Decode-Execute cycle.

Unit 2: The Central Processing Unit (CPU)

The CPU (Central Processing Unit) is the brain of the computer. It fetches instructions from memory, decodes them, executes them, and stores results. CPU performance is measured in GHz (clock speed) and by the number of cores.

Component	Function
ALU (Arithmetic Logic Unit)	Performs all arithmetic (add, subtract) and logical (AND, OR, NOT) operations
CU (Control Unit)	Directs and coordinates the activities of the CPU and other components
Registers	Tiny, ultra-fast storage locations inside the CPU for immediate data
Cache Memory	High-speed memory between CPU and RAM; L1 (fastest), L2, L3 (larger but slower)
Clock Speed	Number of instruction cycles per second; measured in GHz (billions of cycles)
Cores	Independent processing units within a CPU; more cores = better multitasking
Thermal Paste	Compound applied between CPU and heatsink to improve heat transfer

- CPU socket types vary by manufacturer: Intel uses LGA sockets; AMD uses AM4/AM5
- Heat is the CPU's main enemy — always ensure proper cooling (heatsink + fan or liquid cooling)
- A CPU cooler must match the socket type of the motherboard
- Overclocking increases clock speed beyond manufacturer specs but generates more heat

EXAM TIP: The Fetch-Decode-Execute cycle: (1) Fetch — CU gets the next instruction from RAM using the Program Counter. (2) Decode — CU interprets the instruction. (3) Execute — ALU or other unit performs the operation. This repeats billions of times per second.

Unit 3: Memory Systems

Computer memory stores data and instructions for the CPU. Memory is categorised by speed, capacity, and volatility (whether it retains data when power is off).

Memory Type	Full Name	Volatile?	Use
RAM	Random Access Memory	Yes	Temporary working memory; holds running programs and open data
ROM	Read-Only Memory	No	Stores firmware/BIOS; cannot be changed during normal operation
Cache	CPU Cache (L1/L2/L3)	Yes	Ultra-fast buffer between CPU and RAM
HDD	Hard Disk Drive	No	Magnetic storage; large capacity, slower speed
SSD	Solid State Drive	No	Flash storage; faster, more durable, quieter than HDD
Flash Memory	USB drives, SD cards	No	Portable, removable storage
Virtual Memory	Swap file on HDD/SSD	No	Used when RAM is full; much slower than actual RAM

RAM Standards

- DDR4 — current mainstream standard; speeds from 2133 MHz to 3200 MHz+
- DDR5 — next-generation; higher bandwidth, lower voltage
- RAM must match motherboard specifications (type, speed, voltage)
- Dual-channel configuration uses two matching RAM sticks for higher bandwidth

EXAM TIP: RAM is volatile — all data is lost when power is turned off. ROM is non-volatile. Know the memory hierarchy: Registers > Cache > RAM > HDD/SSD, where speed decreases and capacity increases as you go down.

Unit 4: Motherboard & Input/Output Devices

The motherboard is the main printed circuit board (PCB) that connects and facilitates communication between all computer components. It is the backbone of the entire system.

Component	Description
Chipset	Controls data flow between CPU, RAM, and peripheral devices; North and South bridge
BIOS/UEFI	Firmware stored in ROM; initialises hardware during boot (POST); allows hardware settings
PCIe Slot	High-speed expansion slot for GPU, NVMe SSDs, and add-in cards
SATA Port	Connects HDDs and optical drives to the motherboard
USB Headers	Internal connectors for front-panel USB ports
Power Connector	24-pin ATX connector provides main power to the motherboard
Form Factor	Physical size standard: ATX (standard), Micro-ATX, Mini-ITX (smallest)

Common I/O Ports

- USB (Universal Serial Bus) — most common; connects keyboards, mice, drives, phones
- HDMI / DisplayPort — video and audio output to monitors and displays
- Ethernet (RJ-45) — wired network connection
- 3.5mm Audio Jack — headphones and microphones
- VGA — older analogue video port; still found on projectors and older monitors

EXAM TIP: POST (Power-On Self-Test) runs every time the computer starts. It tests CPU, RAM, and basic hardware. If POST fails, the computer beeps in a specific pattern (beep code) indicating the faulty component.

Unit 5: Storage Devices & Power Supply

Storage devices permanently retain data. The Power Supply Unit (PSU) converts mains AC electricity into the DC voltages required by computer components.

Term	Definition
HDD (Hard Disk Drive)	Magnetic storage using spinning platters and read/write heads; high capacity, lower cost
SSD (Solid State Drive)	Flash-based storage; faster, silent, shock-resistant, but more expensive per GB
NVMe SSD	Next-gen SSD using PCIe interface; significantly faster than SATA SSD
Optical Drive	CD/DVD/Blu-ray drive; reads and writes optical discs
PSU (Power Supply Unit)	Converts 230V AC to 3.3V, 5V, and 12V DC for computer components
Wattage	The total power a PSU can supply; must exceed total system power consumption
80+ Rating	PSU efficiency standard; 80+ Bronze, Silver, Gold, Platinum, Titanium

PSU Connectors

- 24-pin ATX — main motherboard power
- 4/8-pin EPS — additional CPU power connector
- 6/8-pin PCIe — power for discrete graphics cards
- SATA power — powers SATA drives and some case fans
- Molex — older 4-pin connector for older drives and fans

EXAM TIP: A PSU should have 20–30% headroom above estimated system power draw for stability and longevity. A failing PSU causes random crashes, BSODs, and system instability — it is often misdiagnosed as a software issue.

Unit 6: Computer Maintenance & Troubleshooting

Regular maintenance prevents hardware failures and extends the life of computer components. Troubleshooting follows a systematic process to identify and resolve hardware and software problems.

Preventive Maintenance Tasks

- Clean dust from fans, heatsinks, and vents using compressed air (every 3–6 months)
- Check and reseat RAM, GPU, and cables if instability occurs
- Monitor CPU and GPU temperatures using software tools (e.g., HWMonitor, SpeedFan)
- Replace thermal paste on CPU every 2–3 years for optimal thermal conductivity
- Backup data regularly to external drives or cloud storage
- Run disk health checks using S.M.A.R.T. data (for HDDs) or CrystalDiskInfo
- Update drivers and firmware to fix bugs and improve compatibility

Troubleshooting Steps (CompTIA A+ Method)

1. Identify the problem — gather information from the user; observe symptoms
2. Establish a theory — consider likely causes based on symptoms
3. Test the theory — verify or eliminate the probable cause
4. Create an action plan — decide on the fix and obtain necessary tools/parts
5. Implement the solution — carry out the repair
6. Verify functionality — confirm the problem is resolved; test thoroughly
7. Document the finding — record the problem, cause, and solution for future reference

EXAM TIP: Know the difference between preventive maintenance (done regularly to avoid failure) and corrective maintenance (done after a failure occurs). ESD (Electrostatic Discharge) can permanently damage components — always use an anti-static wrist strap when handling internal parts.

Unit 7: Networks & Connectivity Basics

A computer network connects two or more computers to share resources and information. Understanding basic networking is essential for hardware technicians.

Term	Definition
LAN	Local Area Network — connects devices in a small area (home, office)
WAN	Wide Area Network — connects devices over large geographic areas; the Internet is a WAN
Router	Directs data packets between networks; connects a LAN to the Internet
Switch	Connects multiple devices within a LAN; smarter than a hub
NIC	Network Interface Card — hardware that connects a computer to a network
IP Address	Unique numerical address assigned to each device on a network
MAC Address	Unique hardware identifier burned into every NIC; format: XX:XX:XX:XX:XX:XX
DHCP	Dynamic Host Configuration Protocol — automatically assigns IP addresses to devices

Network Cables

- Cat5e — supports up to 1 Gbps over 100 metres
- Cat6 — supports up to 10 Gbps over shorter distances; better shielding
- Cat6a — 10 Gbps over 100 metres; used in professional installations
- Fibre optic — uses light pulses; very high speed and long distances; immune to interference

EXAM TIP: Know the Wi-Fi standards: 802.11n (Wi-Fi 4), 802.11ac (Wi-Fi 5), 802.11ax (Wi-Fi 6). Higher numbers mean faster speeds and better range. For wired connections, Ethernet is always more reliable and faster than Wi-Fi.

Unit 8: Operating Systems & Software Installation

The Operating System (OS) is system software that manages computer hardware, provides a user interface, and runs application programs. A hardware technician must be able to install and configure operating systems.

Term	Definition
Operating System	Software that manages hardware resources and provides services to applications
BIOS/UEFI	Initialises hardware and loads the OS bootloader during startup
Boot sequence	The order in which the BIOS checks storage devices to find an OS
Partition	A logical division of a physical storage device
File system	Method of organising data on a storage device (NTFS, FAT32, ext4)
Driver	Software that allows the OS to communicate with specific hardware
Device Manager	Windows tool for viewing, updating, and troubleshooting hardware drivers

OS Installation Steps

- 1. Set the BIOS boot order to boot from USB or DVD
- 2. Boot from the installation media
- 3. Select language, time zone, and keyboard layout
- 4. Choose installation type (Clean install or Upgrade)
- 5. Select and format the target partition
- 6. Wait for file copying and installation to complete
- 7. Configure user account, updates, and network settings
- 8. Install drivers for all hardware components

EXAM TIP: NTFS (Windows) supports files larger than 4GB and has security permissions. FAT32 has a 4GB file size limit. exFAT is used for flash drives as it supports large files without the overhead of NTFS. Linux uses ext4 as its primary file system.